DMMST — Version 2: full results (notebooks 01–06)

Contents

1 Summary **2**

2 Data and protocol **2**
- 2.1 Datasets . . . 2
- 2.2 Protocol . . . 2
- 2.3 Metrics . . . 3
- 2.4 Models . . . 3

3 Tuning (notebook 01) **3**

4 Benchmark (notebooks 02, 03) **4**
- 4.1 C_{td} at each time point (SurvTRACE table format) . . . 7

5 Survival losses (notebook 05) **9**

6 Input encoding and representation (notebook 04) **12**

7 Regression head (notebook 06) **13**

8 Notes on the protocol **15**

9 Conclusion **16**

1 Summary

We tested the components of the DMMST paper — the numeric input embedding (§2.2), the extra survival losses L_{rank} and L_{mul} (§2.3), and the regression ("time of event") head with L_{MAE} and L_{MM} (§2.4) — on five public datasets, and compared the model with eight published survival models under one protocol.

- **Accuracy.** Our model is competitive but not the best: it is in the upper half on every dataset and ties SurvTRACE on hsa synthetic, but wins no dataset (C_{td} 0.000–0.018 behind the best model).
- **Losses.** L_{rank} gives small C_{td} gains on every dataset (significant only on hsa synthetic). The main effect of L_{rank} and L_{mul} is reliability: they remove failed splits and, on EBMT, badly calibrated ones.
- **Regression head.** The best-guess loss (Eq. 6) ranks predicted times better than MAE on observed events only, on every dataset. The head can be trained together with the survival head without harming it when L_{rank} is used; on hsa synthetic this also improves the predicted times, on DeepHit synthetic it does not.
- **Encoding.** The paper's numeric embedding beats discretisation on METABRIC but not on SUPPORT. How the transformer outputs are pooled matters little, except that [CLS] is worst.

2 Data and protocol

2.1 Datasets

Dataset	Type	Patients	Events	Source
METABRIC	real	1,904	1 (death)	DeepSurv (Katzman et al. 2018)
SUPPORT	real	8,873	1 (death)	DeepSurv (Katzman et al. 2018)
EBMT	real	2,279	4 (recovery, adverse event, relapse, death)	R package `mstate`
hsa synthetic	synthetic	5,000	2, both can occur	Tjandra et al. (2021)
DeepHit synthetic	synthetic	30,000	2, competing	Lee et al. (2018)

SEER. SEER was planned as an additional large real dataset (it is used by SurvTRACE). Because of the updated SEER+ data-access policies we could not obtain access to it, so it is not part of these results.

2.2 Protocol

The protocol follows SurvTRACE and DSM:

- each dataset is split 10 times at random into 60% train, 10% validation and 30% test; every model sees the same splits; results are mean (standard deviation) over the 10 splits;
- all settings are chosen on the validation split of split 0 (notebook 01); the test splits are used only for the final numbers;
- every model is scored by the same code; censoring weights and Kaplan–Meier "best guesses" are computed from the training patients of each split only.

2.3 Metrics

- C_{td}: time-dependent concordance (IPCW), the mean over the 25/50/75% quantiles of the event times and over events. Higher is better; 0.5 is random.
- Brier score at the same time points, and the integrated Brier score (IBS) and integrated binomial log-likelihood (IBLL) over the follow-up. Lower is better.
- Cumulative/dynamic AUC and Antolini's C (higher is better).
- D-calibration p-value (higher means no evidence of miscalibration).
- For predicted times: MAE (margin and pseudo-observation versions, in the dataset's time unit) and Harrell's C of the predicted time.

2.4 Models

Our model is a transformer with the paper's numeric embedding and survival head (piecewise-constant hazard). Loss recipes of the survival head:

S1	L_{PCH}
S2	$L_{PCH} + L_{rank}$ (Eq. 4)
S3	$L_{PCH} + L_{mul}$ (Eq. 5) — multi-event data only
S4	$L_{PCH} + L_{rank} + L_{mul}$ — multi-event data only

Recipes of the regression head:

R1	L_{MAE} on observed events only
R2	L_{MAE} with the Kaplan–Meier best guess for censored patients (Eq. 6)
R3	R2 + L_{MM} (Eq. 7), t_q = observed time — multi-event data only
R4	R2 + L_{MM}, t_q = best guess — multi-event data only

L_{mul} and L_{MM} compare different events of the same patient, so they are not defined on single-event data ("—" in the tables). The sign of L_{rank} and L_{mul} in the paper (Eqs. 4–5) is reversed; we use the correct direction, as in DeepHit.

Baselines. SurvTRACE (authors' released code), DeepHit, DSM and MENSA (on our tuned transformer backbone), Random Survival Forest, PC-Hazard, DeepSurv and Cox. Single-event baselines are fitted per event (cause-specific) on multi-event data.

3 Tuning (notebook 01)

Models were tuned on validation C_{td} (24 configurations sampled from SurvTRACE's grid of 216 for our model and SurvTRACE; full grids for Cox and RSF; 16 of 36 for DeepSurv and PC-Hazard). Loss weights were then tuned with the chosen backbone.

Table 1: Chosen settings of our model and of SurvTRACE (ST). lr = learning rate, wd = weight decay.

Dataset	lr	wd	layers	hidden	interm.	heads	ST lr	ST layers	ST hidden
METABRIC	0.001	0	2	8	32	1	0.0001	4	8
SUPPORT	0.001	0.001	3	8	64	2	0.0001	4	8
EBMT	0.001	0.001	3	8	64	2	0.0001	4	16
hsa synth.	0.001	0.001	3	16	64	1	0.001	3	8
DeepHit synth.	0.001	0.001	3	16	64	2	0.001	3	16

Table 2: Chosen loss settings. The raw-day time unit is used only by regression-only models; models with both heads always measure time in units of the end of follow-up (Section 7).

Dataset	L_{rank} weight / σ	L_{mul} weight / σ	L_{MM} weight	time unit (regression only)
METABRIC	1 / 1	—	—	end of follow-up
SUPPORT	0.5 / 0.5	—	—	raw days
EBMT	1 / 0.5	0.5 / 0.5	0.5	raw days
hsa synth.	1 / 0.5	0.1 / 1	0.1	raw days
DeepHit synth.	1 / 0.5	0.5 / 0.5	0.1	raw days

4 Benchmark (notebooks 02, 03)

Our model in this section is recipe S1 (L_{PCH} only); the other recipes are compared in Section 5. Bold is the best per dataset.

Table 3: C_{td} (higher is better).

Model	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
Ours (S1)	0.667 (0.013)	0.622 (0.007)	0.528 (0.008)	0.860 (0.010)	0.735 (0.008)
SurvTRACE	**0.678** (0.013)	**0.638** (0.008)	0.546 (0.009)	**0.868** (0.008)	0.755 (0.007)
DeepHit	0.677 (0.012)	0.610 (0.008)	0.525 (0.011)	0.865 (0.010)	0.737 (0.006)
RSF	0.672 (0.014)	0.636 (0.007)	**0.549** (0.009)	0.849 (0.009)	0.733 (0.005)
PC-Hazard	0.668 (0.017)	0.627 (0.008)	0.548 (0.010)	0.837 (0.008)	0.762 (0.003)
DeepSurv	0.640 (0.030)	0.593 (0.007)	0.529 (0.014)	0.842 (0.009)	**0.762** (0.002)
MENSA	0.654 (0.017)	0.596 (0.021)	0.501 (0.019)	0.847 (0.014)	0.570 (0.074)
Cox	0.631 (0.022)	0.567 (0.006)	0.531 (0.014)	0.497 (0.013)	0.591 (0.005)
DSM	0.620 (0.065)	0.602 (0.037)	0.512 (0.013)	0.522 (0.031)	0.653 (0.104)

Table 4: Integrated Brier score (lower is better).

Model	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
Ours (S1)	0.179 (0.013)	0.199 (0.004)	0.254 (0.099)	0.074 (0.002)	0.179 (0.009)
SurvTRACE	0.201 (0.033)	**0.193** (0.004)	0.242 (0.018)	**0.071** (0.003)	0.169 (0.006)
DeepHit	0.198 (0.007)	0.217 (0.004)	0.388 (0.159)	0.091 (0.002)	0.254 (0.012)
RSF	0.176 (0.007)	0.193 (0.003)	0.224 (0.009)	0.087 (0.002)	0.185 (0.005)
PC-Hazard	**0.174** (0.006)	0.194 (0.004)	0.219 (0.010)	0.078 (0.003)	0.171 (0.005)
DeepSurv	0.176 (0.006)	0.195 (0.004)	0.220 (0.008)	0.078 (0.003)	**0.167** (0.005)
MENSA	0.189 (0.012)	0.212 (0.014)	0.331 (0.023)	0.079 (0.003)	0.223 (0.014)
Cox	0.175 (0.006)	0.198 (0.004)	**0.218** (0.009)	0.105 (0.002)	0.218 (0.006)
DSM	0.224 (0.067)	0.205 (0.008)	0.326 (0.024)	0.369 (0.049)	0.232 (0.078)

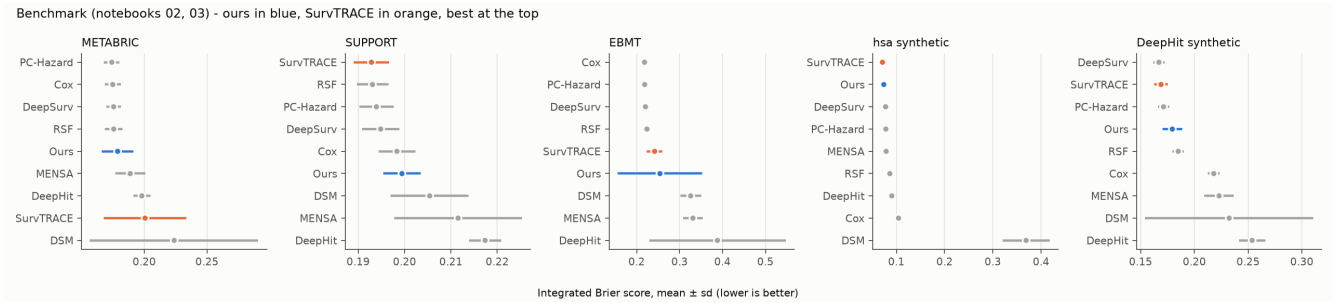


Figure 1: Integrated Brier score per dataset; ours in blue, SurvTRACE in orange.

Table 5: Brier score at the 25/50/75% time points (lower is better).

Model	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
Ours (S1)	0.180 (0.008)	0.197 (0.002)	0.184 (0.089)	0.085 (0.003)	0.129 (0.005)
SurvTRACE	0.216 (0.048)	0.193 (0.003)	0.175 (0.019)	**0.082** (0.003)	0.119 (0.003)
DeepHit	0.190 (0.005)	0.217 (0.004)	0.338 (0.199)	0.105 (0.002)	0.156 (0.004)
RSF	**0.178** (0.005)	**0.192** (0.002)	0.152 (0.004)	0.100 (0.002)	0.133 (0.002)
PC-Hazard	0.178 (0.005)	0.194 (0.002)	0.152 (0.003)	0.090 (0.003)	0.118 (0.002)
DeepSurv	0.179 (0.006)	0.198 (0.003)	0.151 (0.004)	0.089 (0.003)	**0.116** (0.002)
MENSA	0.193 (0.009)	0.209 (0.012)	0.318 (0.021)	0.090 (0.005)	0.170 (0.026)
Cox	0.179 (0.004)	0.201 (0.002)	**0.151** (0.004)	0.121 (0.002)	0.157 (0.002)
DSM	0.247 (0.113)	0.205 (0.008)	0.281 (0.019)	0.382 (0.042)	0.219 (0.152)

Table 6: Cumulative/dynamic AUC (higher is better).

Model	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
Ours (S1)	0.693 (0.021)	0.643 (0.008)	0.526 (0.012)	0.877 (0.011)	0.787 (0.009)
SurvTRACE	**0.707** (0.016)	**0.664** (0.008)	0.545 (0.013)	**0.885** (0.009)	0.805 (0.007)
DeepHit	0.706 (0.015)	0.634 (0.010)	0.524 (0.012)	0.882 (0.010)	0.790 (0.007)
RSF	0.698 (0.017)	0.663 (0.009)	**0.549** (0.011)	0.866 (0.009)	0.781 (0.006)
PC-Hazard	0.695 (0.019)	0.653 (0.010)	0.547 (0.011)	0.853 (0.008)	0.811 (0.003)
DeepSurv	0.670 (0.032)	0.620 (0.008)	0.533 (0.016)	0.858 (0.009)	**0.812** (0.003)
MENSA	0.679 (0.023)	0.611 (0.031)	0.499 (0.020)	0.863 (0.014)	0.574 (0.088)
Cox	0.661 (0.023)	0.595 (0.007)	0.535 (0.015)	0.498 (0.013)	0.584 (0.005)
DSM	0.641 (0.073)	0.621 (0.044)	0.512 (0.015)	0.524 (0.032)	0.684 (0.130)

Table 7: Antolini's C (higher is better).

Model	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
Ours (S1)	0.653 (0.020)	0.610 (0.008)	0.526 (0.009)	0.801 (0.008)	0.724 (0.008)
SurvTRACE	0.660 (0.017)	**0.632** (0.007)	**0.541** (0.005)	**0.811** (0.009)	0.741 (0.007)
DeepHit	**0.662** (0.016)	0.612 (0.007)	0.527 (0.008)	0.806 (0.008)	0.725 (0.006)
RSF	0.656 (0.012)	0.631 (0.008)	0.537 (0.006)	0.787 (0.007)	0.719 (0.005)
PC-Hazard	0.653 (0.010)	0.627 (0.008)	0.536 (0.005)	0.776 (0.009)	0.748 (0.003)
DeepSurv	0.642 (0.018)	0.609 (0.008)	0.531 (0.006)	0.782 (0.008)	**0.749** (0.003)
MENSA	0.639 (0.025)	0.581 (0.034)	0.501 (0.011)	0.788 (0.013)	0.564 (0.069)
Cox	0.637 (0.015)	0.595 (0.006)	0.537 (0.011)	0.500 (0.007)	0.583 (0.004)
DSM	0.614 (0.054)	0.597 (0.036)	0.504 (0.014)	0.516 (0.028)	0.646 (0.099)

Table 8: Integrated binomial log-likelihood (lower is better).

Model	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
Ours (S1)	0.528 (0.033)	0.583 (0.012)	0.731 (0.300)	0.243 (0.007)	0.536 (0.025)
SurvTRACE	0.585 (0.082)	**0.568** (0.011)	0.679 (0.041)	**0.235** (0.007)	0.507 (0.017)
DeepHit	0.578 (0.018)	0.625 (0.009)	1.788 (1.419)	0.290 (0.004)	0.700 (0.030)
RSF	0.519 (0.018)	0.568 (0.009)	0.645 (0.027)	0.287 (0.005)	0.551 (0.015)
PC-Hazard	**0.516** (0.016)	0.571 (0.011)	0.628 (0.028)	0.257 (0.007)	0.513 (0.015)
DeepSurv	0.522 (0.017)	0.571 (0.010)	0.632 (0.021)	0.257 (0.008)	**0.498** (0.015)
MENSA	0.571 (0.052)	0.616 (0.044)	0.878 (0.057)	0.259 (0.011)	0.646 (0.040)
Cox	0.520 (0.017)	0.579 (0.010)	**0.627** (0.025)	0.349 (0.006)	0.628 (0.016)
DSM	0.641 (0.173)	0.596 (0.021)	0.862 (0.060)	0.949 (0.121)	0.948 (1.039)

Table 9: D-calibration p-value (higher = no evidence of miscalibration).

Model	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
Ours (S1)	0.250 (0.300)	0.042 (0.108)	0.119 (0.144)	0.307 (0.258)	0.228 (0.158)
SurvTRACE	0.000 (0.000)	0.001 (0.001)	0.000 (0.000)	0.423 (0.284)	0.281 (0.284)
DeepHit	0.000 (0.000)	0.000 (0.000)	0.010 (0.021)	0.000 (0.000)	0.000 (0.000)
RSF	0.510 (0.296)	0.021 (0.040)	0.346 (0.129)	0.006 (0.009)	**0.533** (0.144)
PC-Hazard	0.330 (0.345)	0.022 (0.039)	0.231 (0.090)	0.742 (0.164)	0.135 (0.124)
DeepSurv	**0.652** (0.231)	0.027 (0.024)	0.393 (0.092)	0.585 (0.191)	0.181 (0.122)
MENSA	0.006 (0.012)	0.002 (0.007)	0.000 (0.000)	0.089 (0.070)	0.105 (0.173)
Cox	0.629 (0.305)	**0.075** (0.094)	**0.407** (0.113)	**0.893** (0.107)	0.293 (0.208)
DSM	0.022 (0.070)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.240 (0.260)

Table 10: MAE of the expected time read from the survival curve (margin MAE, linear tail; dataset time unit; lower is better).

Model	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
Ours (S1)	90.8 (5.2)	722.2 (16.1)	5316.1 (406.8)	1176.1 (197.8)	39.8 (1.3)
SurvTRACE	**80.2** (3.6)	**708.3** (15.8)	4855.4 (119.2)	17885.7 (50752.1)	36.5 (0.8)
DeepHit	118.7 (14.8)	808.1 (20.1)	5751.6 (1655.4)	750.7 (91.4)	174.1 (16.9)
RSF	85.3 (2.4)	748.2 (10.4)	19956938.7 (33630886.7)	**384.3** (7.7)	44.7 (0.4)
PC-Hazard	96.7 (7.1)	711.3 (20.5)	5400.8 (234.7)	1173.0 (92.2)	36.9 (0.6)
DeepSurv	81.4 (3.2)	716.1 (9.0)	5879.1 (358.3)	1267.1 (103.1)	**34.7** (0.3)
MENSA	107.2 (50.9)	742.3 (25.4)	**4595.0** (82.1)	1701.7 (584.7)	45.7 (1.8)
Cox	81.4 (2.2)	731.2 (8.0)	5616.6 (297.1)	449.0 (9.0)	45.1 (0.3)
DSM	102.7 (19.4)	763.4 (29.2)	4632.0 (111.1)	466.3 (26.9)	58.9 (37.7)

4.1 C_{td} at each time point (SurvTRACE table format)

Table 11: METABRIC.

Model	25%	50%	75%
Ours (S1)	0.695 (0.017)	0.666 (0.014)	0.642 (0.020)
SurvTRACE	0.701 (0.021)	**0.678** (0.018)	**0.655** (0.013)
DeepHit	**0.707** (0.014)	0.675 (0.016)	0.648 (0.012)
RSF	0.705 (0.016)	0.668 (0.017)	0.644 (0.016)
PC-Hazard	0.691 (0.019)	0.668 (0.020)	0.644 (0.021)
DeepSurv	0.638 (0.041)	0.639 (0.034)	0.643 (0.019)
MENSA	0.679 (0.026)	0.651 (0.018)	0.633 (0.018)
Cox	0.628 (0.033)	0.628 (0.025)	0.637 (0.015)
DSM	0.629 (0.086)	0.621 (0.058)	0.610 (0.051)

Table 12: SUPPORT.

Model	25%	50%	75%
Ours (S1)	0.651 (0.011)	0.613 (0.008)	0.601 (0.007)
SurvTRACE	**0.666** (0.012)	**0.631** (0.009)	0.617 (0.005)
DeepHit	0.640 (0.009)	0.600 (0.010)	0.591 (0.008)
RSF	0.662 (0.008)	0.629 (0.009)	**0.618** (0.006)
PC-Hazard	0.651 (0.009)	0.620 (0.011)	0.609 (0.007)
DeepSurv	0.586 (0.008)	0.588 (0.007)	0.605 (0.007)
MENSA	0.623 (0.022)	0.588 (0.021)	0.577 (0.024)
Cox	0.549 (0.009)	0.563 (0.007)	0.588 (0.006)
DSM	0.629 (0.047)	0.592 (0.034)	0.585 (0.030)

Table 13: EBMT, per event.

	recovery			adverse event			relapse			death		
Model	25%	50%	75%	25%	50%	75%	25%	50%	75%	25%	50%	75%
Ours (S1)	0.558 (0.025)	0.569 (0.017)	0.560 (0.017)	0.569 (0.032)	0.516 (0.022)	0.511 (0.022)	0.474 (0.038)	0.488 (0.019)	0.494 (0.028)	0.515 (0.035)	0.544 (0.027)	0.532 (0.026)
SurvTRACE	0.564 (0.013)	0.573 (0.011)	0.570 (0.010)	0.596 (0.026)	0.540 (0.019)	**0.529** (0.019)	0.513 (0.049)	0.508 (0.028)	**0.516** (0.028)	0.527 (0.030)	0.562 (0.022)	0.552 (0.020)
DeepHit	0.566 (0.029)	0.570 (0.022)	0.571 (0.018)	0.530 (0.072)	0.497 (0.038)	0.499 (0.029)	0.505 (0.047)	0.508 (0.031)	0.503 (0.039)	0.509 (0.028)	0.529 (0.035)	0.518 (0.031)
RSF	0.550 (0.018)	0.575 (0.017)	0.567 (0.013)	**0.597** (0.026)	**0.544** (0.014)	0.529 (0.017)	**0.555** (0.027)	**0.522** (0.022)	0.504 (0.032)	0.525 (0.029)	**0.569** (0.025)	**0.552** (0.017)
PC-Hazard	0.563 (0.014)	0.576 (0.013)	0.568 (0.013)	0.581 (0.031)	0.537 (0.015)	0.523 (0.015)	0.554 (0.043)	0.521 (0.024)	0.503 (0.027)	0.532 (0.040)	0.564 (0.019)	0.552 (0.018)
DeepSurv	0.560 (0.012)	0.558 (0.016)	0.563 (0.014)	0.530 (0.035)	0.516 (0.017)	0.522 (0.018)	0.486 (0.057)	0.488 (0.039)	0.478 (0.022)	0.541 (0.036)	0.560 (0.027)	0.551 (0.012)
MENSA	0.491 (0.041)	0.500 (0.045)	0.491 (0.044)	0.515 (0.069)	0.519 (0.026)	0.509 (0.018)	0.479 (0.050)	0.494 (0.032)	0.497 (0.023)	0.500 (0.025)	0.507 (0.036)	0.507 (0.019)
Cox	**0.576** (0.016)	**0.579** (0.015)	**0.577** (0.016)	0.505 (0.034)	0.501 (0.022)	0.510 (0.022)	0.490 (0.063)	0.486 (0.031)	0.496 (0.037)	**0.542** (0.037)	0.562 (0.025)	0.550 (0.015)
DSM	0.509 (0.043)	0.503 (0.043)	0.508 (0.044)	0.552 (0.064)	0.517 (0.030)	0.499 (0.023)	0.534 (0.061)	0.498 (0.024)	0.497 (0.040)	0.498 (0.031)	0.491 (0.044)	0.536 (0.022)

Table 14: hsa synthetic, per event.

	event 1			event 2		
Model	25%	50%	75%	25%	50%	75%
Ours (S1)	0.897 (0.011)	0.861 (0.013)	0.831 (0.011)	0.902 (0.015)	0.848 (0.013)	0.822 (0.015)
SurvTRACE	**0.904** (0.016)	**0.867** (0.015)	**0.836** (0.015)	**0.910** (0.015)	**0.859** (0.011)	**0.834** (0.010)
DeepHit	0.900 (0.016)	0.864 (0.012)	0.833 (0.012)	0.907 (0.015)	0.856 (0.010)	0.830 (0.009)
RSF	0.886 (0.011)	0.850 (0.009)	0.818 (0.009)	0.889 (0.019)	0.841 (0.011)	0.811 (0.010)
PC-Hazard	0.870 (0.012)	0.836 (0.012)	0.803 (0.011)	0.884 (0.016)	0.827 (0.012)	0.803 (0.012)
DeepSurv	0.879 (0.009)	0.844 (0.011)	0.813 (0.010)	0.881 (0.016)	0.828 (0.012)	0.803 (0.011)
MENSA	0.885 (0.011)	0.848 (0.017)	0.814 (0.018)	0.893 (0.020)	0.836 (0.019)	0.808 (0.017)
Cox	0.494 (0.020)	0.492 (0.017)	0.494 (0.013)	0.489 (0.024)	0.508 (0.016)	0.508 (0.014)
DSM	0.503 (0.061)	0.508 (0.054)	0.534 (0.035)	0.536 (0.062)	0.530 (0.040)	0.523 (0.048)

Table 15: DeepHit synthetic, per event.

	event 1			event 2		
Model	25%	50%	75%	25%	50%	75%
Ours (S1)	0.768 (0.008)	0.734 (0.009)	0.704 (0.009)	0.770 (0.013)	0.729 (0.010)	0.703 (0.009)
SurvTRACE	0.791 (0.010)	0.754 (0.009)	0.719 (0.007)	0.794 (0.012)	0.750 (0.006)	0.720 (0.008)
DeepHit	0.771 (0.008)	0.738 (0.009)	0.707 (0.007)	0.771 (0.010)	0.732 (0.008)	0.704 (0.006)
RSF	0.767 (0.005)	0.735 (0.008)	0.702 (0.009)	0.765 (0.009)	0.730 (0.008)	0.700 (0.008)
PC-Hazard	0.800 (0.007)	0.761 (0.006)	0.725 (0.005)	**0.802** (0.009)	0.757 (0.005)	0.726 (0.006)
DeepSurv	**0.802** (0.007)	**0.763** (0.006)	**0.726** (0.005)	0.802 (0.009)	**0.757** (0.004)	**0.726** (0.005)
MENSA	0.576 (0.094)	0.563 (0.073)	0.548 (0.061)	0.593 (0.086)	0.575 (0.073)	0.563 (0.065)
Cox	0.612 (0.010)	0.588 (0.007)	0.566 (0.006)	0.623 (0.008)	0.587 (0.008)	0.570 (0.005)
DSM	0.670 (0.121)	0.648 (0.108)	0.629 (0.097)	0.683 (0.116)	0.653 (0.100)	0.634 (0.087)

Observations.

- SurvTRACE is the strongest model on METABRIC, SUPPORT and hsa synthetic; RSF and PC-Hazard on EBMT; DeepSurv and PC-Hazard on DeepHit synthetic. Our model (S1) is in the upper half.
- Our tuned SurvTRACE reproduces its published C_{td} (0.678 vs 0.691 on METABRIC, 0.638 vs 0.640 on SUPPORT), which supports the correctness of the protocol.
- Our calibration (IBS) is close to the best on four datasets and better than SurvTRACE's on METABRIC; on EBMT our IBS is high and varies between splits.
- DeepHit, DSM and MENSA are poorly calibrated (D-calibration $p \approx 0$) and DSM and MENSA are unstable across splits.
- EBMT is hard for every model (C_{td} 0.50–0.55): its four features are categorical and weak.

5 Survival losses (notebook 05)

Table 16: C_{td} of each loss recipe.

Recipe	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
S1: L_{PCH}	0.667 (0.013)	0.615 (0.011)	0.529 (0.013)	0.849 (0.049)	0.743 (0.003)
S2: $+L_{rank}$	**0.670** (0.016)	**0.620** (0.007)	0.535 (0.016)	**0.868** (0.010)	**0.743** (0.004)
S3: $+L_{mul}$	—	—	0.530 (0.016)	0.859 (0.015)	0.742 (0.003)
S4: $+L_{rank} + L_{mul}$	—	—	**0.537** (0.015)	0.867 (0.009)	0.742 (0.004)

Table 17: Integrated Brier score of each loss recipe.

Recipe	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
S1: L_{PCH}	0.179 (0.012)	0.200 (0.004)	0.338 (0.165)	0.075 (0.007)	0.174 (0.005)
S2: $+L_{rank}$	**0.175** (0.006)	**0.199** (0.004)	0.313 (0.153)	0.075 (0.003)	**0.171** (0.005)
S3: $+L_{mul}$	—	—	0.256 (0.082)	**0.074** (0.003)	0.175 (0.005)
S4: $+L_{rank} + L_{mul}$	—	—	**0.233** (0.043)	0.075 (0.003)	0.172 (0.006)

Table 18: Brier score at the 25/50/75% time points.

Recipe	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
S1: L_{PCH}	0.181 (0.009)	0.198 (0.002)	0.294 (0.192)	0.086 (0.008)	0.125 (0.003)
S2: $+L_{rank}$	**0.179** (0.008)	**0.197** (0.003)	0.256 (0.172)	**0.085** (0.003)	**0.122** (0.002)
S3: $+L_{mul}$	—	—	0.196 (0.093)	0.085 (0.004)	0.125 (0.002)
S4: $+L_{rank} + L_{mul}$	—	—	**0.163** (0.025)	0.086 (0.004)	0.122 (0.002)

Table 19: Cumulative/dynamic AUC.

Recipe	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
S1: L_{PCH}	0.694 (0.019)	0.637 (0.011)	0.533 (0.014)	0.866 (0.052)	0.794 (0.005)
S2: $+L_{rank}$	**0.699** (0.019)	**0.641** (0.010)	0.530 (0.015)	**0.885** (0.010)	**0.795** (0.005)
S3: $+L_{mul}$	—	—	0.526 (0.013)	0.876 (0.015)	0.793 (0.004)
S4: $+L_{rank} + L_{mul}$	—	—	**0.535** (0.011)	0.884 (0.010)	0.794 (0.004)

Table 20: Each recipe against S1 on the same 10 splits: mean difference, number of splits where the recipe's value is higher, and paired p-values.

Metric	Dataset	Recipe	Δ vs S1	wins	p (t-test)	p (Wilcoxon)
C_{td}	DeepHit synth.	S2	+0.0004	5/10	0.762	0.770
C_{td}	DeepHit synth.	S3	-0.0009	5/10	0.548	0.695
C_{td}	DeepHit synth.	S4	-0.0003	4/10	0.779	0.770
C_{td}	EBMT	S2	+0.0060	5/10	0.449	0.846
C_{td}	EBMT	S3	+0.0006	4/10	0.919	0.695
C_{td}	EBMT	S4	+0.0080	7/10	0.345	0.432
C_{td}	hsa synth.	S2	+0.0191	7/10	0.224	0.027
C_{td}	hsa synth.	S3	+0.0102	5/10	0.537	0.770
C_{td}	hsa synth.	S4	+0.0181	7/10	0.254	0.049
C_{td}	METABRIC	S2	+0.0035	5/10	0.473	0.557
C_{td}	SUPPORT	S2	+0.0051	5/10	0.097	0.275
IBS	DeepHit synth.	S2	-0.0037	1/10	0.006	0.006
IBS	DeepHit synth.	S3	+0.0009	6/10	0.547	0.492
IBS	DeepHit synth.	S4	-0.0026	3/10	0.026	0.049
IBS	EBMT	S2	-0.0250	5/10	0.714	0.922
IBS	EBMT	S3	-0.0815	6/10	0.220	0.625
IBS	EBMT	S4	-0.1041	5/10	0.067	0.432
IBS	hsa synth.	S2	-0.0005	7/10	0.788	0.232
IBS	hsa synth.	S3	-0.0006	7/10	0.776	0.375
IBS	hsa synth.	S4	+0.0001	7/10	0.968	0.322
IBS	METABRIC	S2	-0.0043	3/10	0.329	0.432
IBS	SUPPORT	S2	-0.0011	3/10	0.167	0.193

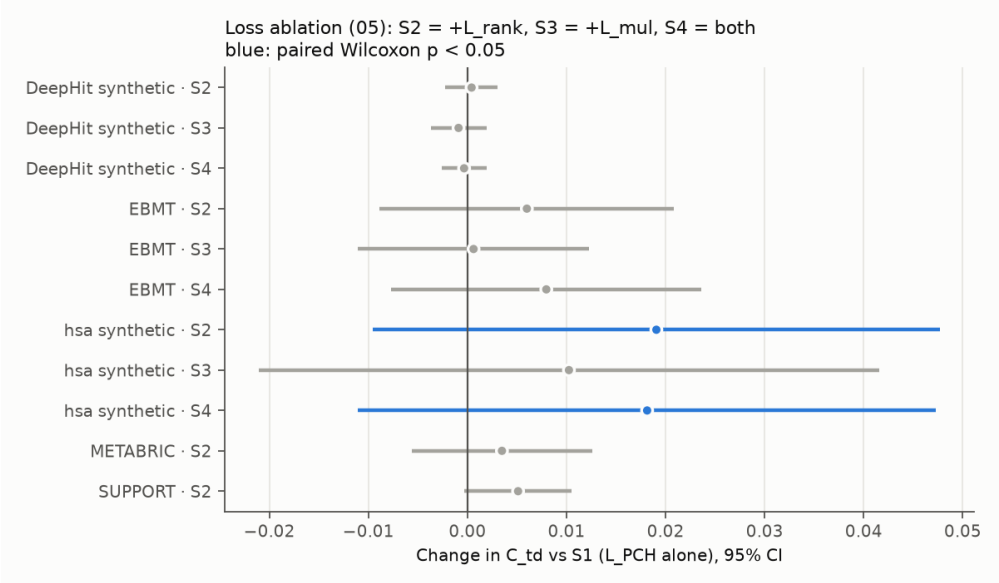


Figure 2: Change in C_{td} against S1 with 95% confidence intervals; blue: paired Wilcoxon $p < 0.05$.

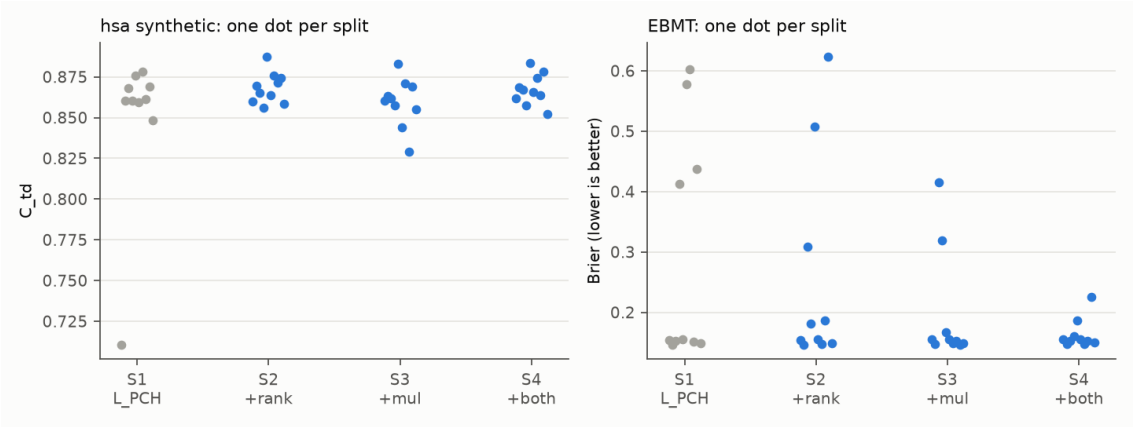


Figure 3: Every split of each recipe. Left: C_{td} on hsa synthetic; right: Brier score on EBMT.

Observations.

- L_{rank} (S2) is at least as good as S1 on every dataset (+0.000 to +0.019 C_{td}); the gain is significant only on hsa synthetic.
- L_{mul} alone (S3) helps less than L_{rank}; S4 is about as good as S2.
- With L_{PCH} alone, one hsa split fails (C_{td} 0.71) and several EBMT splits are badly calibrated (Brier up to 0.6). With $L_{rank} + L_{mul}$ (S4) neither happens: the EBMT IBS drops from 0.338 to 0.233 and its spread from 0.165 to 0.043.
- On DeepHit synthetic all recipes are the same: L_{PCH} alone is enough on this data.

6 Input encoding and representation (notebook 04)

Table 21: Encodings of the numeric inputs and ways of pooling the transformer outputs (over layers / over tokens / which layers).

Variant	METABRIC		SUPPORT	
	C_{td}	IBS	C_{td}	IBS
encoding: numeric embedding (paper)	0.666 (0.023)	0.178 (0.010)	0.617 (0.007)	0.200 (0.004)
encoding: quantile	0.669 (0.014)	**0.174** (0.005)	0.620 (0.006)	0.200 (0.005)
encoding: bins	0.642 (0.019)	0.188 (0.013)	**0.627** (0.006)	**0.195** (0.004)
repr.: sum / max / all layers	0.667 (0.014)	0.179 (0.013)	0.617 (0.009)	0.201 (0.004)
repr.: sum / max / last layers	0.665 (0.013)	0.181 (0.013)	0.619 (0.008)	0.200 (0.005)
repr.: sum / mean / all layers	0.671 (0.011)	0.178 (0.011)	0.618 (0.010)	0.201 (0.006)
repr.: sum / mean / last layers	0.667 (0.024)	0.178 (0.010)	0.610 (0.018)	0.203 (0.007)
repr.: mean / max / all layers	0.657 (0.023)	0.186 (0.015)	0.615 (0.010)	0.201 (0.004)
repr.: mean / max / last layers	0.664 (0.017)	0.181 (0.013)	0.618 (0.010)	0.200 (0.004)
repr.: mean / mean / all layers	0.665 (0.024)	0.182 (0.013)	0.609 (0.024)	0.204 (0.009)
repr.: mean / mean / last layers	0.654 (0.034)	0.182 (0.015)	0.615 (0.011)	0.202 (0.007)
repr.: concat / max / all layers	0.668 (0.021)	0.177 (0.009)	0.621 (0.009)	0.200 (0.005)
repr.: concat / max / last layers	0.651 (0.034)	0.184 (0.015)	0.611 (0.009)	0.203 (0.009)
repr.: concat / mean / all layers	**0.672** (0.014)	0.177 (0.013)	0.616 (0.010)	0.201 (0.005)
repr.: concat / mean / last layers	0.668 (0.016)	0.182 (0.012)	0.615 (0.008)	0.201 (0.003)
repr.: [CLS] / - / all layers	0.658 (0.020)	0.192 (0.012)	0.584 (0.034)	0.209 (0.010)

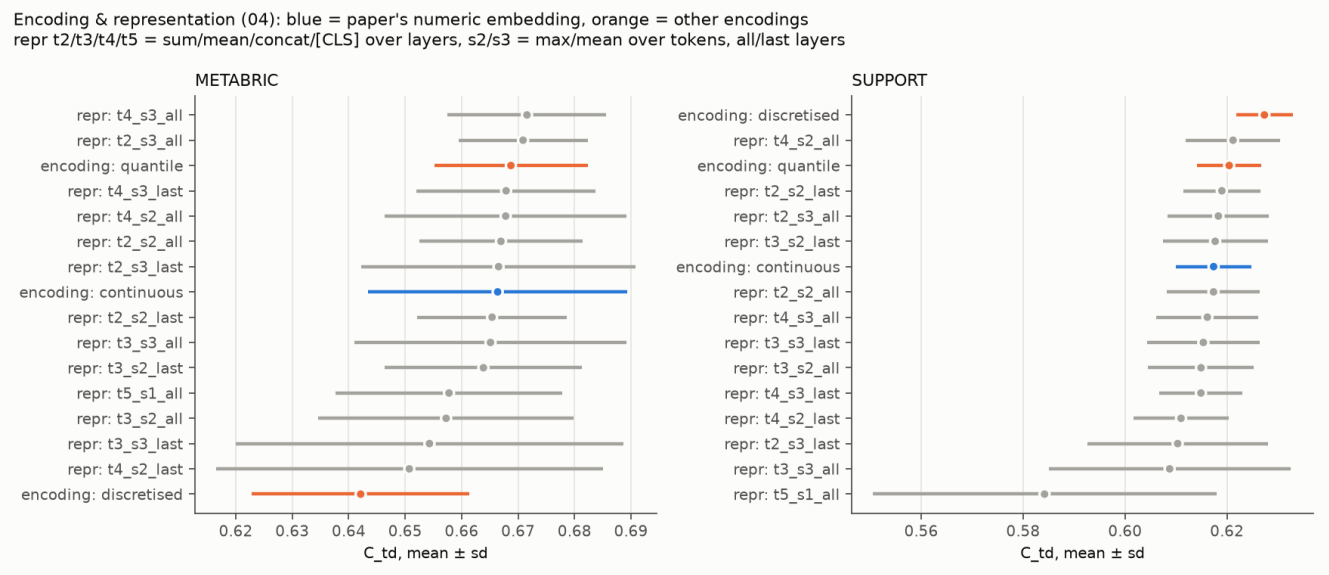


Figure 4: C_{td} of every encoding and representation; blue: the paper's numeric embedding.

Observations.

- On METABRIC the paper's numeric embedding (0.666) is clearly better than discretisation into bins (0.642) and level with quantile scaling. On SUPPORT discretisation is slightly better (0.627 vs 0.617).

- Most ways of pooling the transformer outputs are within ±0.01, inside the split-to-split spread; using the [CLS] token is clearly the worst, especially on SUPPORT.

7 Regression head (notebook 06)

Two setups are trained: the regression head alone (R1–R4), and the regression head together with the survival head ("joint", every S × R). In joint training the time unit is the end of follow-up and early stopping waits 30 validation rounds (Section 8).

Table 22: Predicted times. MAE in the dataset's time unit (lower is better); Harrell's C of the predicted time (higher is better); C_{td} of the survival head for the joint models. "Full (joint)" is the same model as "Ours Full" in the benchmark tables; all joint recipes are in Table 23.

Dataset	Setup	Recipe	MAE-margin	MAE-pseudo-obs.	Harrell C	C_{td}
METABRIC	regression only	R1	80.6 (2.8)	92.5 (3.8)	0.608 (0.018)	—
METABRIC	regression only	R2	76.7 (3.2)	88.6 (4.0)	0.643 (0.012)	—
METABRIC	Full (joint)	S2 × R2	77.6 (3.7)	89.5 (4.2)	0.642 (0.015)	0.672 (0.021)
SUPPORT	regression only	R1	613.9 (9.1)	739.4 (12.7)	0.546 (0.007)	—
SUPPORT	regression only	R2	604.6 (9.6)	730.1 (12.7)	0.592 (0.009)	—
SUPPORT	Full (joint)	S2 × R2	609.5 (15.3)	734.5 (15.3)	0.594 (0.008)	0.619 (0.007)
EBMT	regression only	R1	4254.4 (122.7)	6932.4 (199.3)	0.510 (0.012)	—
EBMT	regression only	R2	4231.9 (130.2)	6909.9 (200.0)	0.524 (0.014)	—
EBMT	regression only	R3	4245.1 (125.6)	6923.1 (202.4)	0.530 (0.009)	—
EBMT	regression only	R4	4247.4 (120.1)	6925.4 (201.8)	0.533 (0.011)	—
EBMT	Full (joint)	S4 × R2	4311.8 (110.8)	6989.8 (185.0)	0.505 (0.018)	0.534 (0.017)
hsa synth.	regression only	R1	396.4 (5.8)	677.8 (9.8)	0.566 (0.021)	—
hsa synth.	regression only	R2	408.7 (96.7)	690.0 (95.0)	0.663 (0.047)	—
hsa synth.	regression only	R3	420.6 (94.9)	702.0 (96.0)	0.663 (0.087)	—
hsa synth.	regression only	R4	379.8 (62.2)	661.1 (63.3)	0.665 (0.063)	—
hsa synth.	Full (joint)	S4 × R2	435.2 (115.8)	531.9 (179.6)	0.767 (0.112)	0.852 (0.041)
DeepHit synth.	regression only	R1	38.8 (1.5)	75.0 (2.3)	0.689 (0.046)	—
DeepHit synth.	regression only	R2	34.8 (3.2)	70.8 (4.2)	0.704 (0.042)	—
DeepHit synth.	regression only	R3	35.0 (3.0)	71.0 (2.9)	0.700 (0.035)	—
DeepHit synth.	regression only	R4	33.9 (0.4)	69.9 (1.8)	0.714 (0.003)	—
DeepHit synth.	Full (joint)	S4 × R2	43.0 (5.2)	79.2 (5.2)	0.576 (0.114)	0.744 (0.004)

Table 23: Joint models on multi-event data: C_{td} / Harrell's C of the predicted time for every survival recipe (rows) and regression recipe (columns).

Dataset	Survival	R1	R2	R3	R4
EBMT	S1	0.529 / 0.506	0.527 / 0.513	0.530 / 0.509	0.525 / 0.498
	S2	0.533 / 0.501	0.530 / 0.493	0.538 / 0.492	0.526 / 0.515
	S3	0.529 / 0.509	0.531 / 0.511	0.533 / 0.506	0.523 / 0.503
	S4	0.530 / 0.493	0.534 / 0.505	0.532 / 0.508	0.539 / 0.506
hsa synth.	S1	0.756 / 0.613	0.750 / 0.675	0.774 / 0.733	0.768 / 0.790
	S2	0.810 / 0.668	0.850 / 0.703	0.847 / 0.705	0.852 / 0.696
	S3	0.719 / 0.615	0.807 / 0.744	0.818 / 0.650	0.749 / 0.688
	S4	0.770 / 0.598	0.852 / 0.767	0.840 / 0.718	0.851 / 0.661
DeepHit synth.	S1	0.743 / 0.589	0.726 / 0.483	0.741 / 0.568	0.743 / 0.584
	S2	0.744 / 0.479	0.744 / 0.529	0.743 / 0.513	0.742 / 0.517
	S3	0.742 / 0.595	0.742 / 0.665	0.742 / 0.571	0.743 / 0.617
	S4	0.744 / 0.544	0.744 / 0.576	0.744 / 0.536	0.743 / 0.457

Table 24: Joint models on single-event data: C_{td} / Harrell's C.

Dataset	Survival	R1	R2
METABRIC	S1	0.671 / 0.616	0.665 / 0.640
	S2	0.671 / 0.605	0.672 / 0.642
SUPPORT	S1	0.620 / 0.551	0.619 / 0.597
	S2	0.621 / 0.556	0.619 / 0.594

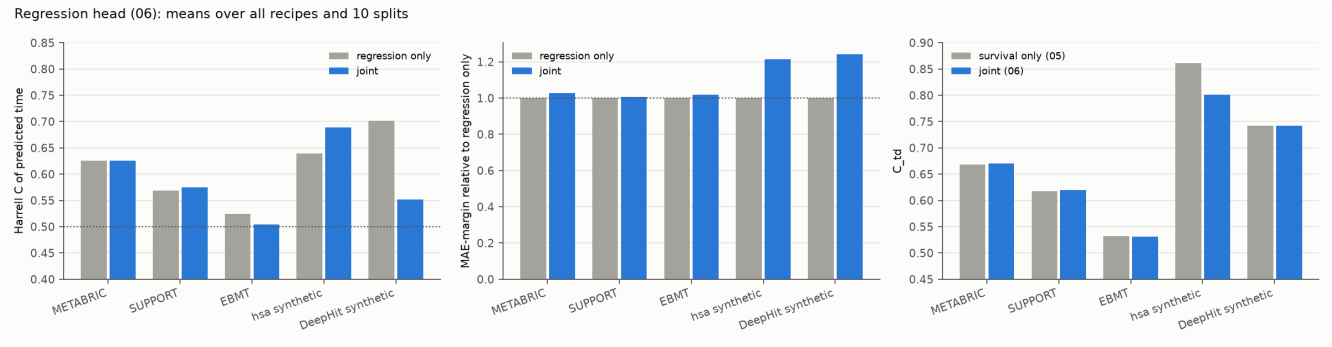


Figure 5: Means over all recipes and splits. Left: Harrell's C of the predicted time; middle: MAE relative to the regression-only models; right: C_{td} of survival-only (notebook 05) and joint models.

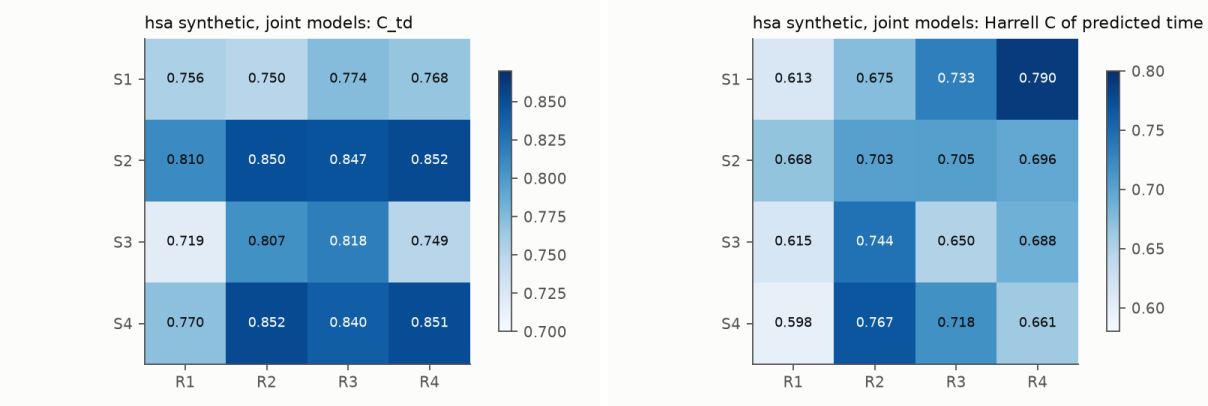


Figure 6: hsa synthetic, joint models: C_{td} (left) and Harrell's C (right) for every recipe.

Observations.

- **Best guess (R2) vs observed events only (R1):** R2 ranks predicted times better on every dataset (Harrell's C, e.g. 0.566 → 0.663 on hsa synthetic, 0.546 → 0.592 on SUPPORT) and has a lower MAE on four of five datasets (not on hsa synthetic).
- L_{MM} **(R3, R4)** changes little on EBMT and DeepHit synthetic; R4 is the best regression-only recipe on both and on hsa synthetic, by small margins.
- **Joint training does not harm the survival head:** C_{td} equals survival-only on METABRIC, SUPPORT, EBMT and DeepHit synthetic. On hsa synthetic it does so only with L_{rank} (S2/S4 with R2–R4: 0.84–0.85 vs 0.86); without L_{rank} (S1/S3) it is lower (0.72–0.82).
- **Effect on the predicted times:** on hsa synthetic the joint models rank times better than the regression head alone (Harrell's C 0.77 for Full, 0.69 on average over all recipes, up to 0.79 for S1 × R4, vs 0.66; the best recipe is picked on test scores, so it is optimistic). On DeepHit synthetic they are worse (Full 0.58, average 0.55, vs 0.70): joint models choose their checkpoint by the survival C_{td}, so the regression head is not what is optimised.
- EBMT: predicted times carry almost no ranking information in any setup (Harrell's $C \approx 0.5$).

8 Notes on the protocol

- **Early stopping.** Joint models can stay near $C_{td} = 0.5$ for the first ∼30 epochs while the two heads settle; with a patience of 10 validation rounds some were stopped there. Joint models and the loss recipes of notebook 05 therefore use a patience of 30. The benchmark runs of notebooks 02–03 use 10; for our model (S1) the two settings agree within 0.011 C_{td}.
- **Time unit.** With raw days the regression loss (hundreds of days) dominated the survival loss in joint training; joint models therefore always use the end of follow-up as time unit.
- **Scoring fixes applied before these results:** the Brier score no longer drops patients censored at the last follow-up time (70% of hsa synthetic), and IBS/IBLL/AUC are computed when a test patient is followed longer than every training patient.
- **Not included:** the language-model experiments (notebook 07, paper §3) and sequential data (paper §4.2, planned for version 3). SEER could not be used (updated SEER+ access policies).

9 Conclusion

Under one protocol on five public datasets, the DMMST model is competitive with strong survival models but does not outperform SurvTRACE in C_{td}. The paper's additional losses are useful mainly for reliability and calibration on multi-event data; the best-guess regression loss is better than using observed events only; and the regression head can be trained together with the survival head without harming it, provided L_{rank} is used.